

Meraj Hossain Promit

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Dhaka, Bangladesh

RESEARCH INTERESTS

Final-year Robotics & Mechatronics undergraduate working toward **robot learning for locomotion across morphologies**. My current research learns recovery policies for a **wheeled** Mars rover trapped in deformable granular terrain: mobility when contact is uncertain and no hand-designed escape recipe exists. I want to carry that problem from **wheels to quadrupeds to bipeds**, where richer contact and stricter balance make model-based controllers fail more sharply. I am drawn to **reinforcement learning, learning-based control, and Sim2Real**, and to putting simulation-trained policies on hardware. Seeking a **PhD / MS / research position** in robot learning.

EDUCATION

University of Dhaka

B.Sc. in Robotics and Mechatronics Engineering;
Fourth Year, Second Semester (8th of 8), Ongoing

Dhaka, Bangladesh
August 2022 – October 2026 (Expected)
CGPA: 3.50 / 4.00

Sylhet MC College

Higher Secondary Certificate (HSC);

Sylhet, Bangladesh
July 2019 – December 2021

PUBLICATIONS

Meraj Hossain Promit, Chandak Chakma, Md. Jubair Ahmed Surov, and Sejuti Rahman. “Escape Without a Recipe: Maneuver-Agnostic Mars Rover Recovery from Granular Entrapment.” *Space Robotics Workshop* at IEEE/RSJ IROS 2026 (Pittsburgh, PA). **Accepted**, poster. [Workshop]

Motivation: Wheel entrapment in loose regolith has crippled real planetary missions (e.g., NASA’s Spirit rover). We learn a maneuver-agnostic recovery policy in deformable granular dynamics, restoring mobility without a prescribed escape recipe.

RESEARCH EXPERIENCE

MARVEL Lab, University of Dhaka

Undergraduate Researcher, Lab Page

Dhaka, Bangladesh

April 2025 – Present

Supervisor: Dr. Sejuti Rahman, Associate Professor (on leave), University of Dhaka; Professor of CS, New Uzbekistan University.

- **Lead author** on DRL controllers that recover rover mobility in deformable Mars regolith, learning locomotion policies where classical control fails under granular slip.
- Designed **yaw-projected reward shaping with entrapment-gated slip penalties** (a safety-style constraint) and built large-scale, multi-GPU training and evaluation pipelines.

University of Texas Rio Grande Valley

Research Assistant

Rio Grande Valley, TX, USA (Remote)

August 2025 – Present

- Building a digital-twin warehouse benchmark to evaluate Deep RL navigation policies (PPO, SAC, TD3) under realistic perception and localization noise. [GitHub]

North South University

Undergraduate Research Student

Dhaka, Bangladesh

January 2024 – December 2024

- Studied melanoma segmentation from pseudo-annotations under limited labels (weakly-supervised computer vision). **Supervised by** Dr. Mahdy Rahman Chowdhury, Associate Professor, ECE, NSU.

PROJECTS

WarehouseBenchmark | Deep RL, NVIDIA Isaac Lab, ROS2

[GitHub]

- Engineered a 31.8×54 m digital-twin warehouse benchmarking PPO, SAC, and TD3 for autonomous navigation on a Clearpath Jackal with realistic LiDAR perception and localization noise.
- Applied **curriculum learning and domain randomization** to close the sim-to-real gap, surfacing how reward design and sensor noise drive policy robustness.

6-DOF Robotic Arm: Sim2Real Manipulation | ROS2, MoveIt, Gazebo, OpenCV

- Transferred a vision-guided pick-and-place pipeline from simulation to physical hardware, with color and shape recognition driving grasp selection.
- Implemented inverse-kinematics solvers and collision-aware trajectory optimization for real-time, closed-loop manipulation.

Custom Quadcopter Flight Controller | STM32, Embedded C, Control Theory

- Built a from-scratch STM32 flight controller with **cascaded PID attitude and rate loops** and IMU sensor fusion, achieving stable autonomous flight.
- Wrote low-level firmware for motor mixing, failsafe protocols, and real-time telemetry.

Autonomous Navigation Stack | ROS2, Nav2, SLAM

- Deployed SLAM mapping, AMCL localization, and hybrid A*/DWA planning for dynamic obstacle avoidance on a real differential-drive robot.

TECHNICAL SKILLS

Reinforcement & Robot Learning: PyTorch, NVIDIA Isaac Lab, PPO / SAC / TD3, reward shaping, domain randomization, curriculum learning, Sim2Real

Robotics & Simulation: ROS2, Gazebo, Nav2, MoveIt, SLAM, OpenCV

Programming & Systems: Python, C / C++ (learning CUDA), Bash; Linux (Ubuntu, Arch, Kali); Git

Hardware & CAD: STM32, Raspberry Pi, Fusion 360, ANSYS, PCB design

LEADERSHIP & ENGINEERING EXPERIENCE

Intelis Solution Ltd.

Dhaka, Bangladesh

Embedded Engineer (Part-time)

July 2026 – Present

- Building STM32-based autonomous drone platforms for maritime monitoring, with robust firmware, hardened telemetry links, and weather-resistant sensing for harsh sea conditions.

Karthikesh Robotics Pvt. Ltd.

Chennai, India (Remote)

ROS Development Intern

June 2025 – Present

- Developing autonomous robots for industrial and educational use with ROS2 navigation and perception stacks.

Mohakashchhari, Dhaka University Rover Team

Dhaka, Bangladesh

Project Lead

May 2025 – Present

- Proposed the concept at IEEE SB DU; now leading design, frame integration, and fundraising for a competition-ready rover platform.

IEEE Robotics & Automation Society, University of Dhaka

Dhaka, Bangladesh

Chair (2025–Present); previously Student Activity Secretary (2023–2024)

- Lead chapter activities, events, and research collaborations promoting robotics and automation among students.

AWARDS & ACHIEVEMENTS

8th Place, DU AI Challenge (2025): computer-vision competition among university teams.

4th Place, Bangladesh Olympiad on Astronomy & Astrophysics (2021): national winner and training-camp participant.

Rank 62, HSC 2021 (Sylhet Board): top 100 across all examinees in the board.

RELEVANT COURSEWORK

Robotics & AI: Artificial Intelligence, Intelligent Systems & Robotics, Advanced Robotics, Control Systems Design, Digital Image Processing & Robot Vision, Digital Signal Processing, Mechatronics (Intro & Advanced)

Math & CS: Linear Algebra, Multivariable & Vector Calculus, Differential Equations, Numerical Analysis, Probability & Statistics, Data Structures & Algorithms, Object-Oriented Programming